SHEET INDEX

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PART TYPES (how each part is identified)

Type	Identified by
B bought	its line in docs/BOM.csv; modelled as an envelope, exact only where it meets other parts
C cut stock	item C-01 to C-11 on D-02 = the rows of cad/out/cut_list.csv (extrusions, lead screw, dowels)
P printed	file name in cad/out/printed/ (<part>.step / .stl); qty, material and print note from print_list.json
W plywood	the bed (D-03)
S sheet metal	the 1 mm aluminium flags (D-06); probe flag outline cad/out/probe_flag_blank.dxf

GENERAL NOTES

1. Units: millimetres; angles in degrees. Do not scale the drawing.

2. Projection: third angle. Views are hidden-line-removed projections of cad/model.py (build.py poses; printed parts as exported); hidden edges, where shown, are dashed; a section view hatches the faces its cutting plane cuts (D-17).

3. Coordinates (cad/layout.py): X left to right along the pocket row, Y front to back, Z up; Z 0 = top of the pocket pads. Cup X = machine X of the suction-cup axis; c = height of the cup face. Datum O (X 0, Y 0): D-03.

4. Every dimension is measured from the CAD by cad/drawings.py or read from a named cad/layout.py constant; cad/out/drawings/dims.json lists each one with its source, and a self-check re-measures them from fresh geometry (0.05). Confidence tags [Certain] / [Likely] / [Guessing] are the docs'. The title block's SOURCE hash is the last commit that changed cad/model.py, layout.py, dims.py or drawings.py ('+uncommitted': the working tree differs from it).

5. Bought parts are envelopes; check them on arrival (BUILD_GUIDE 3.3). Not in the CAD and not drawn: frame joints (BUILD_GUIDE 4.3), cables and drag chain, vacuum tube, belt slack, supports of the light roof.

6. GENERAL TOLERANCES, unless a sheet states otherwise:

   a) Extrusion cut lengths +/-0.5, square ends (BUILD_GUIDE 4.1, the supplier's cut tolerance). Phase 0 jig posts 312 +/-0.5, square (BUILD_GUIDE 1.1). Lead screw: 311 to 356 below the motor face works (cut_list.csv).

   b) Frame (BUILD_GUIDE 4.5, D-02): base-rail centres +/-0.5; upright inner faces +/-0.5 [Certain, CAD]; bed top to beam bottom +/-0.5 [Certain, CAD]; diagonals equal within 1. X rail top edge 4.00 +/-0.1 below the beam top (6.2).

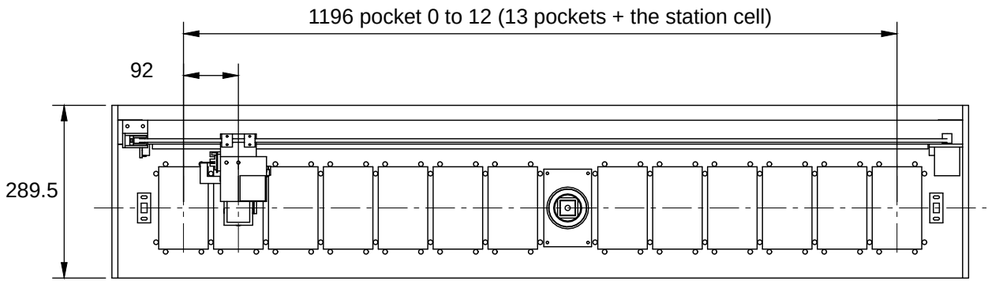
   c) Set on assembly (BUILD_GUIDE): spine bottom 275 +/-1 above the pad top (7.1); Z carriage bottom 295 +/-2 above the cup face (7.5); X home click at cup X 19 +/-2 (6.8); Z top click at c 258 +/-2 (7.4); carry-sensor trip at c 246.5 +/-1 (7.4a); probe trip at 2 +/-0.5 of buffer travel (7.6); station lens front 23 +/-1 above the pads (5.9); laser beam 214 +/-1 above the pad top (12.5); flag bend 90 +/-0.5° (7.6); Phase 0 jig: slab ledge 140 +/-2 above the station lens (1.5).

   d) PROPOSED, no doc states one (to agree in review): printed parts +/-0.3; plywood outline +/-0.5; sheet-metal outlines +/-0.2; dowel cut length +/-0.5. Printed clearance holes are drilled out after printing: M3 3.4, M5 5.5 (BUILD_GUIDE 3.4).

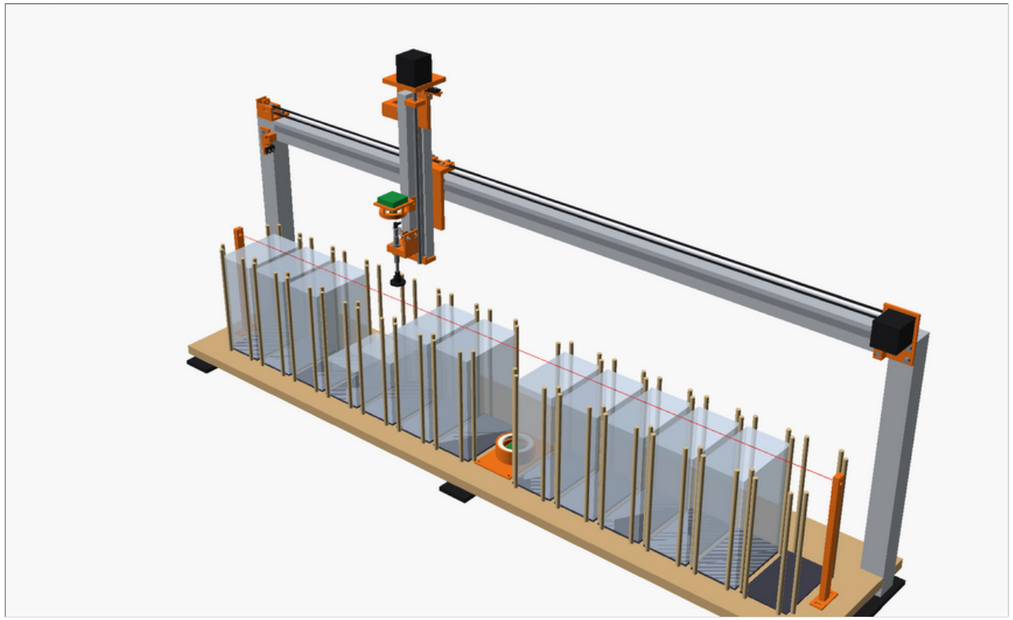
7. Status: Rev A, draft for design review. Nothing has been built; the Phase 0 go / no-go tests (D-17) come first. Not released for fabrication.

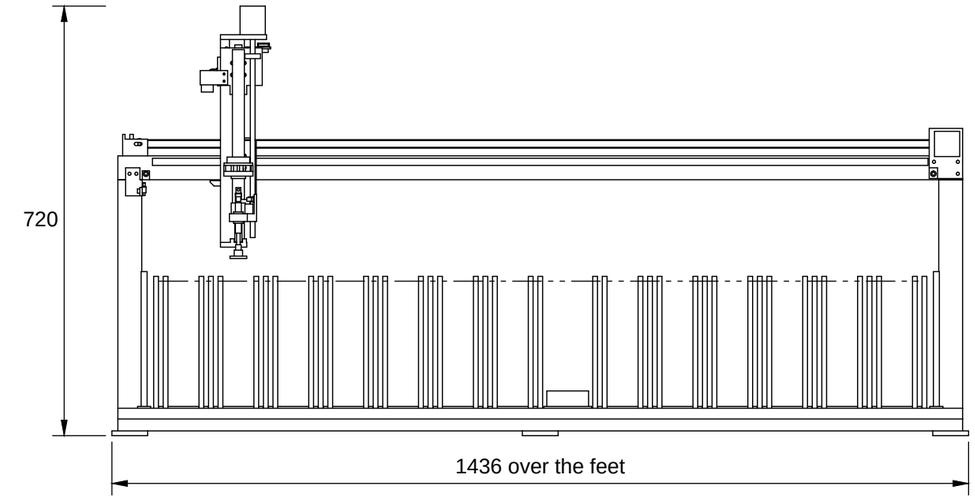
SlabRail - graded-slab batch scanner	**SHEET D-00 of D-17**
Index and general notes	SCALE NONE (text sheet)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

TOP



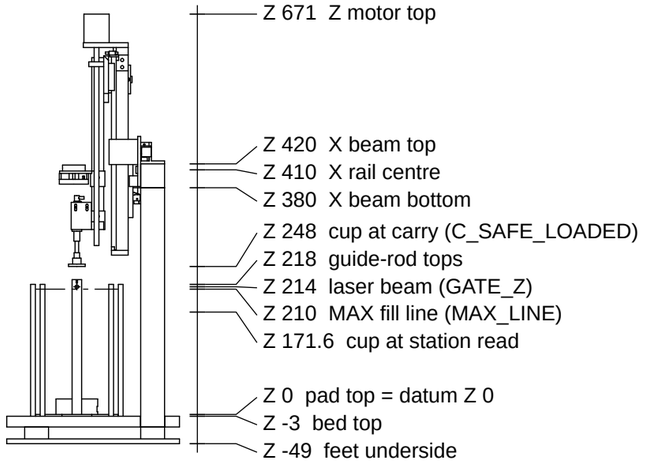
ISO (render, docs/img/machine_iso.png; not to scale)





FRONT (looking along +Y)

Heights, machine Z (cup face where noted)

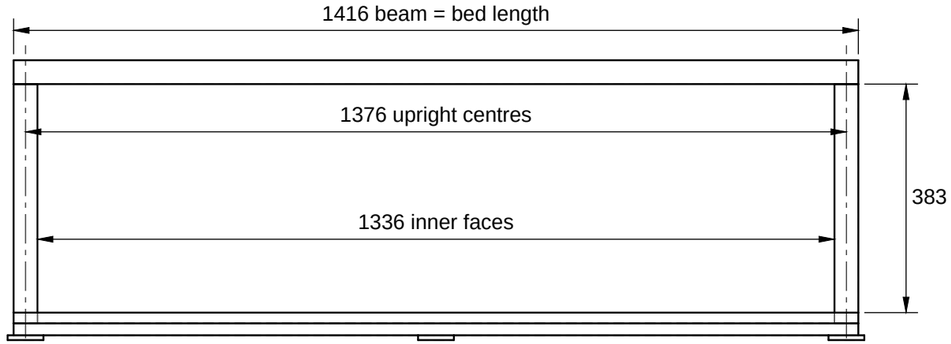


RIGHT SIDE (along -X)

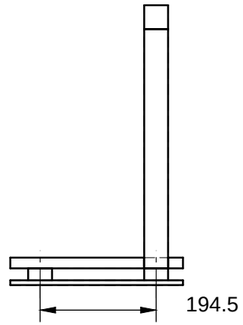
Main data	Value	From
Pockets, one row	13 + station (cell 7)	layout
Pitch; slabs per pocket	92; 30	layout
Pocket interior X x Y	84 x 139	layout
Guide rods	82 x Ø8	CAD
X used (cup X)	42 to 1238	CAD
X stops; home trip	15 / 1262; 19	layout
Z used (cup face c)	-12 to 250	layout
Z hard stops (c)	-14 / 262	layout

Pose 'home' (build.POSES): cup over pocket 1 at the carry height. Not shown: fasteners, slabs, light roof, cables. Heights are machine Z above the pad top (Z 0); X from the machine origin O (D-03). Dash-dot: MAX fill line; cross on the right view: the overfill laser beam.

SlabRail - graded-slab batch scanner	**SHEET D-01 of D-17**
General arrangement	SCALE 1:10
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**



FRAME, FRONT (1:10)



RIGHT (1:10)

Frame set-up (BUILD_GUIDE 4.5; tolerances quoted)

Base rails: centre distance (RIGHT view) +/-0.5 at both ends; diagonals equal within 1. Upright inner faces (FRONT) +/-0.5, equal at the bottom and the top [Certain, CAD]. Bed top to beam bottom +/-0.5, equal at both ends [Certain, CAD]. Uprights stand on the back base rail through notches in the bed (D-03); the beam sits on the uprights, flush with their outer faces. Joints (L brackets, back joining plates) are NOT in the CAD: BUILD_GUIDE 4.3 [Guessing on the exact layout]. Nor are the bed-to-base-rail bolts (16x M5x25 with washers into T-nuts in the rails' outer slots, BUILD_GUIDE 4.3 [Likely]): drill the bed for them on assembly, after a test on a rail offcut.

CUT LIST (cad/out/cut_list.csv; extrusion cuts +/-0.5 with square ends, BUILD_GUIDE 4.1)

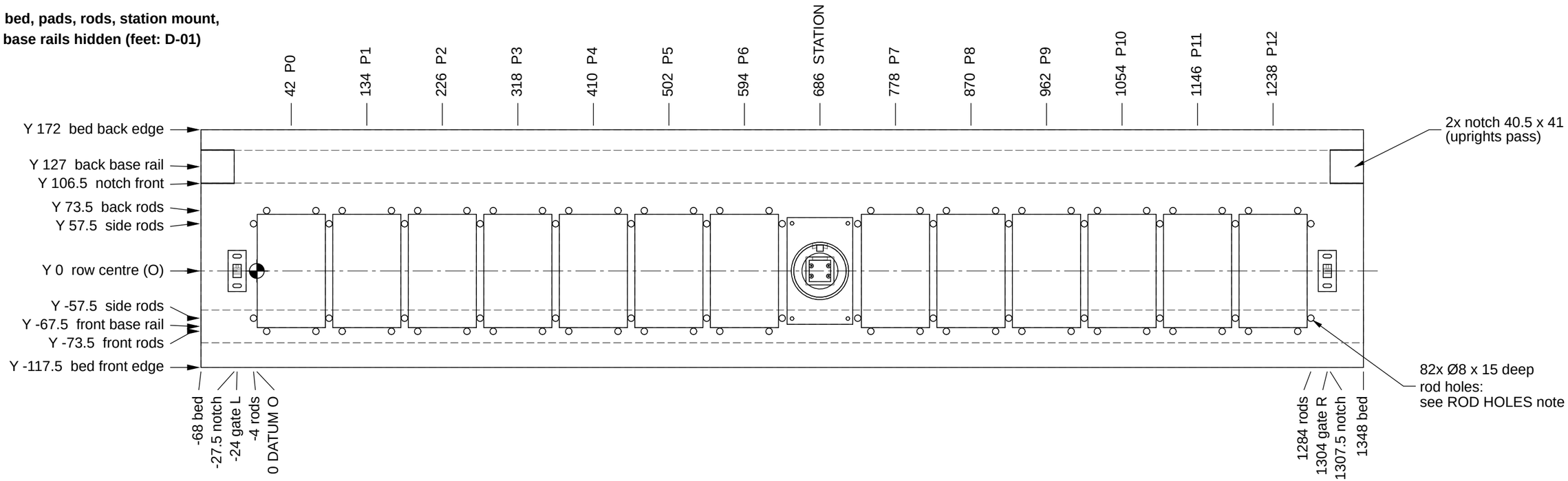
Extrusions: 20-series, slot 6 (Misumi HFS5, BOM.csv)

Item	Bar (length from the CAD; scale per row)	Profile / stock	Use, ends, holes	Qty	Tolerance
C-01	1416 — 1:10	2040	base rail, lying flat under the bed; square ends	2	+/-0.5
C-02	401 — 1:10	4040	upright; square ends	2	+/-0.5
C-03	1416 — 1:10	4040	X beam, on top of the uprights; square ends	1	+/-0.5
C-04	340 — 1:5	2040	Z spine; BOTH ends: 2x M5 tapped in the two centre bores (20 apart), full thread ≥ 8 deep. Top end: 2x M5x16 through z_motor_mount, 8 engaged. Bottom end: 2x M5x12 button heads through z_end_cap, 7 engaged (BUILD_GUIDE 7.1, 7.3)	1	+/-0.5
C-05	285 — 1:5	2020	head drop arm; 2x Ø5.5 THRU across the arm (machine X), 19 and 84 from the end in head_foot's socket: match-drill (BUILD_GUIDE 7.5)	1	+/-0.5
C-06	340 — 1:5	T8x8	T8x8 Z screw, length below the motor face: cut the motor's 400 mm screw; 311-356 works (cut_list.csv)	1	311-356
C-07		1 mm Al sheet	probe flag blank: D-06	1	D-06
C-08	312 — 1:5	2020	Phase 0 jig post; cut square	2	+/-0.5
C-09	66 — 1:5	2020	Phase 0 jig arm stub; Ø5.5 THRU cross hole 19 from one end, match-drilled through head_foot (BUILD_GUIDE 1.5)	1	+/-0.5
C-10		1 mm Al sheet	carry-sensor flag: D-06	1	D-06
C-11	236 — 1:5	5/16 in dowel	guide rod; dome the top R2-3 (ROD_DOME_R; BUILD_GUIDE 5.4), chamfer the bottom lightly	82	PROPOSED +/-0.5

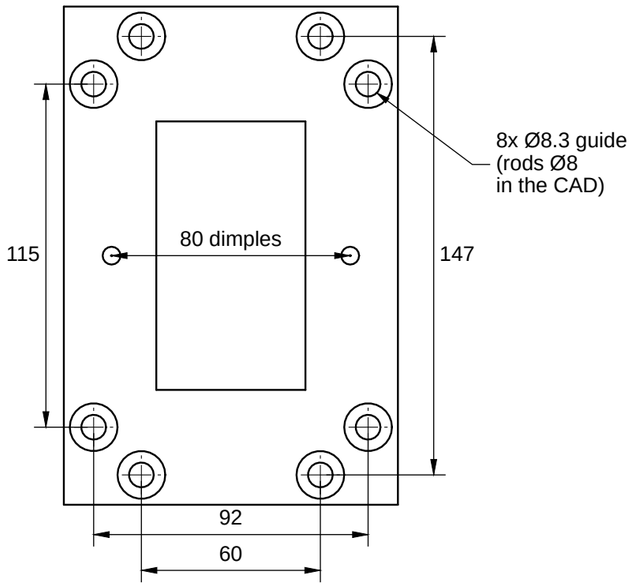
Not in cut_list.csv: the optional rear diagonal brace, 2020 x 600 (BOM.csv HFS5-2020-600), cut only if the beam-twist check fails (BUILD_GUIDE 6.7).

SlabRail - graded-slab batch scanner	**SHEET D-02 of D-17**
Frame and extrusion cut list	SCALE 1:5, 1:10 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

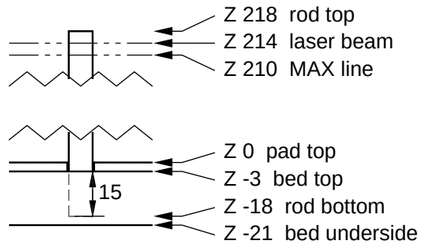
PLAN (1:5): bed, pads, rods, station mount, gate posts; base rails hidden (feet: D-01)



DETAIL C (1:2): one cell's rod holes = drill_template
(cell centre at 0, 0; the machine drills them, BUILD_GUIDE 5.6)



DETAIL D (1:2): rod in the bed,
front, broken



Bed and pads

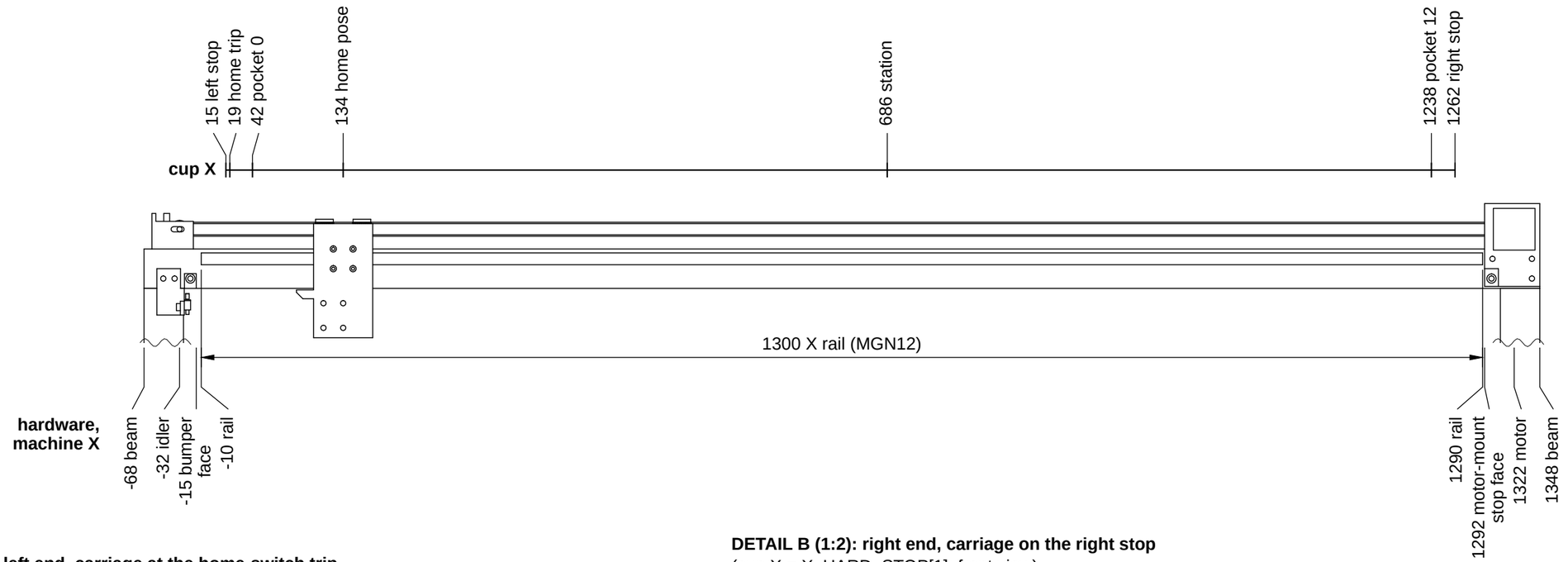
Item	Size (from the CAD)	Note
Bed, plywood (W)	1416 x 289.5 x 18	cut on site; test-drill for voids first (BUILD_GUIDE 5.1)
Pads (EVA + microfiber), 13	83 x 138 x 3	spray adhesive, after the rods (5.9)
Rods (dowels), 82	D-02 item C-11	glued; plumb within 1 over 200 [Guessing target] (5.7)

Datum O = machine origin (X 0; Y 0 = the pocket-row centre line), as the software uses it; X ordinates from O. The machine drills the rod holes through drill_template, so they are not laid out from the bed edges (BUILD_GUIDE 5.6). Station mount: cell 7 (X 686), four #8 wood screws, no pad. Gate posts: bare plywood, no shim (5.9).

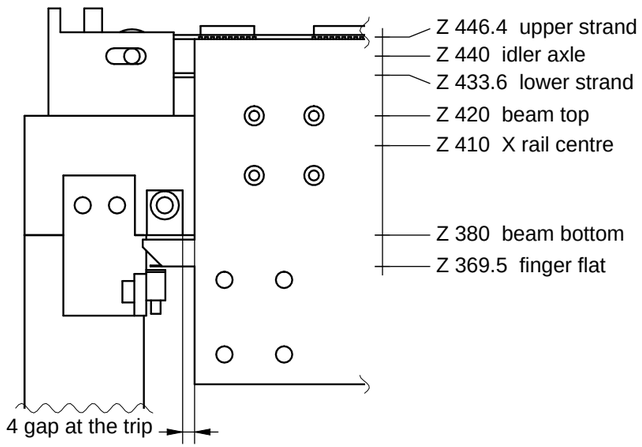
ROD HOLES: 82x Ø8 (ROD_D; brad-point bit to suit the dowels: 5/16 in for the BOM's 7.94 dowels) x 15 deep (ROD_EMBED), square to the bed (guide block, 5.6). Side rods on the wall lines: X -4, then every 92, at Y ±57.5. Front/back rods: pairs 60 apart about each pocket centre, at Y ±73.5; none in cell 7.

SlabRail - graded-slab batch scanner	**SHEET D-03 of D-17**
Bed, pockets and guide rods	SCALE 1:2, 1:5 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

FRONT (1:5)

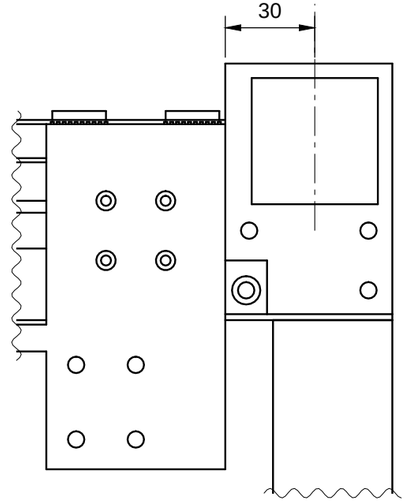


DETAIL A (1:2): left end, carriage at the home-switch trip
(cup X = X_HOME_SWITCH; front view)



Click at cup X 19 (+/-2, BUILD_GUIDE 6.8)

DETAIL B (1:2): right end, carriage on the right stop
(cup X = X_HARD_STOP[1]; front view)



x_bracket lands on the mount's left end

Y stack-up, cup axis to the X beam's front face

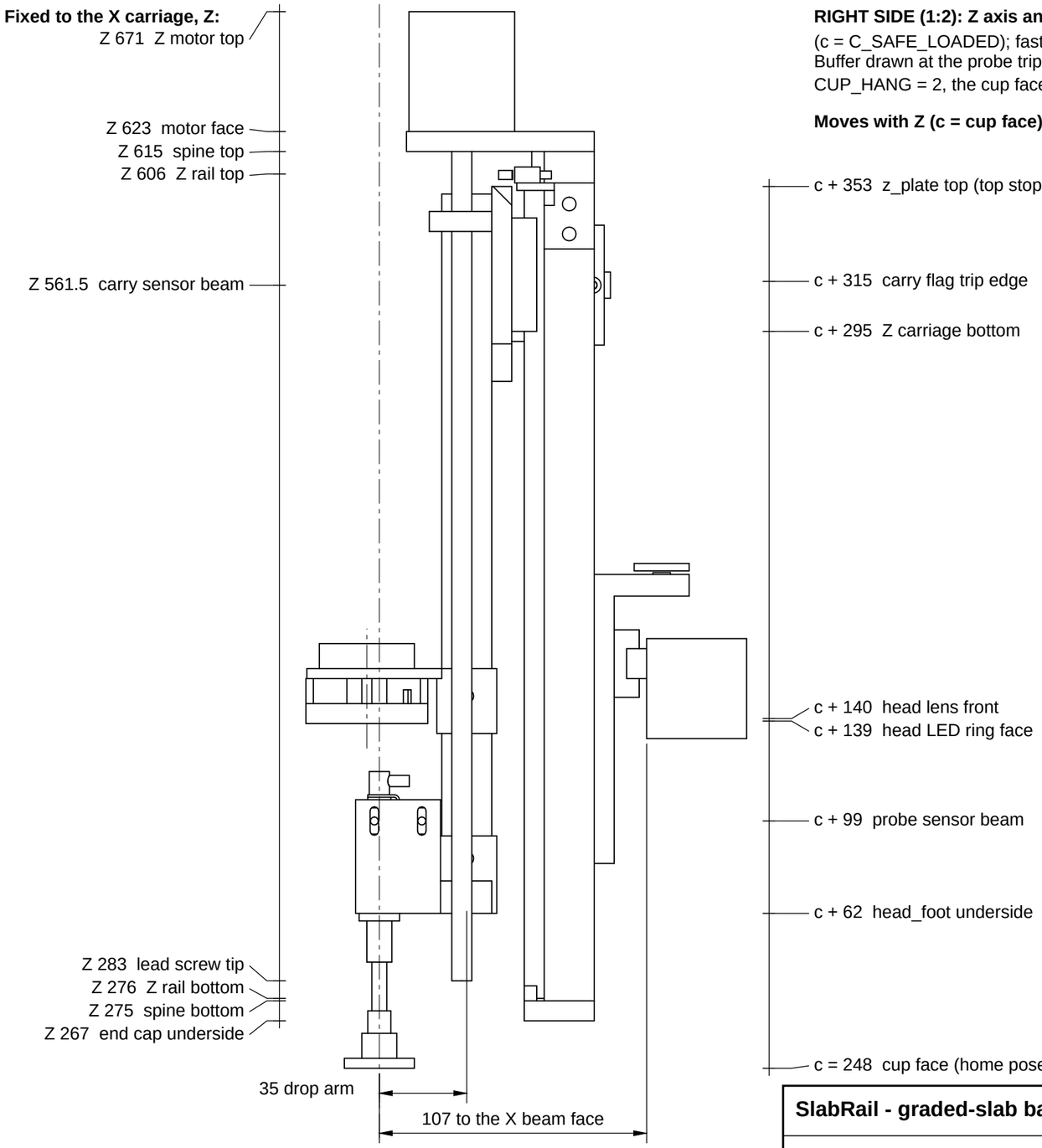
Constant (layout.py)	mm	Part
ARM_TO_CUP	35	drop-arm centre to cup axis
EXT20 / 2	10 (of 20)	drop arm (2020): half = 10
Z_PLATE_T	8	z_plate
MGN12_H	13	Z carriage (MGN12H)
EXT20	20	Z spine (2040, 20 deep)
X_BRACKET_T	8	x_bracket
MGN12_H	13	X carriage (MGN12H)
= Y_CUP_TO_BEAM	107	sum: cup axis to beam front face

CAD check: beam front face at Y 107

FRONT: X axis with the carriage at home (cup over pocket 1); Z axis and head not shown. cup X = machine X of the suction-cup axis. Belt: GT2 loop over the beam top, both ends clamped on the x_bracket tab (BUILD_GUIDE 6.5).

SlabRail - graded-slab batch scanner	**SHEET D-04 of D-17**
X axis	SCALE 1:2, 1:5 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

Fixed to the X carriage, Z:



RIGHT SIDE (1:2): Z axis and head, home pose

(c = C_SAFE_LOADED); fasteners not shown
Buffer drawn at the probe trip: compressed
CUP_HANG = 2, the cup face at c.

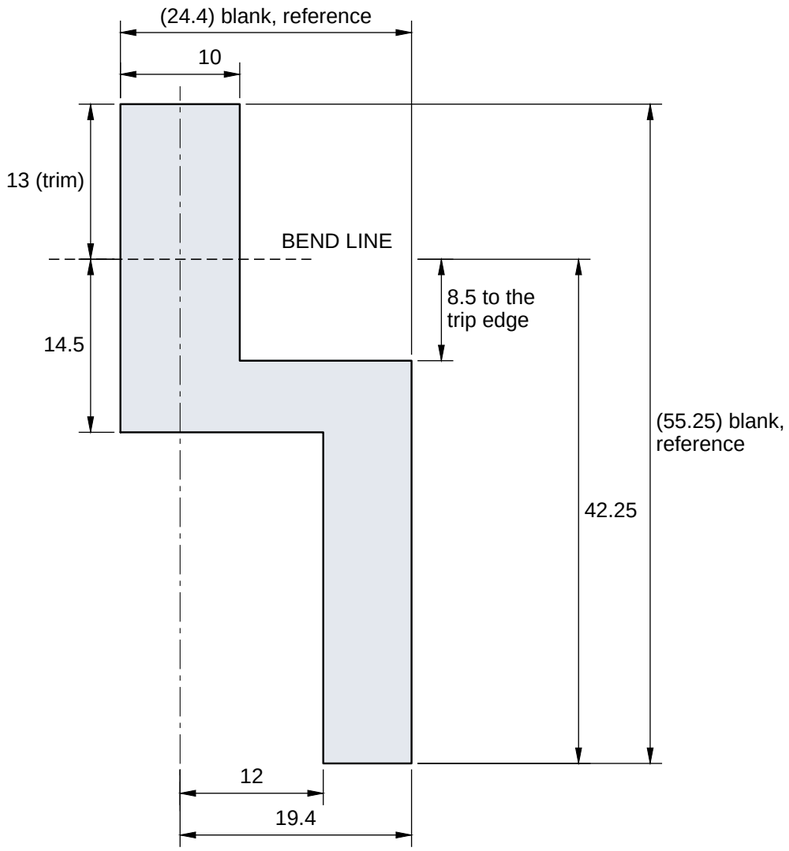
Moves with Z (c = cup face):

Z-height table (values read from cad/layout.py)

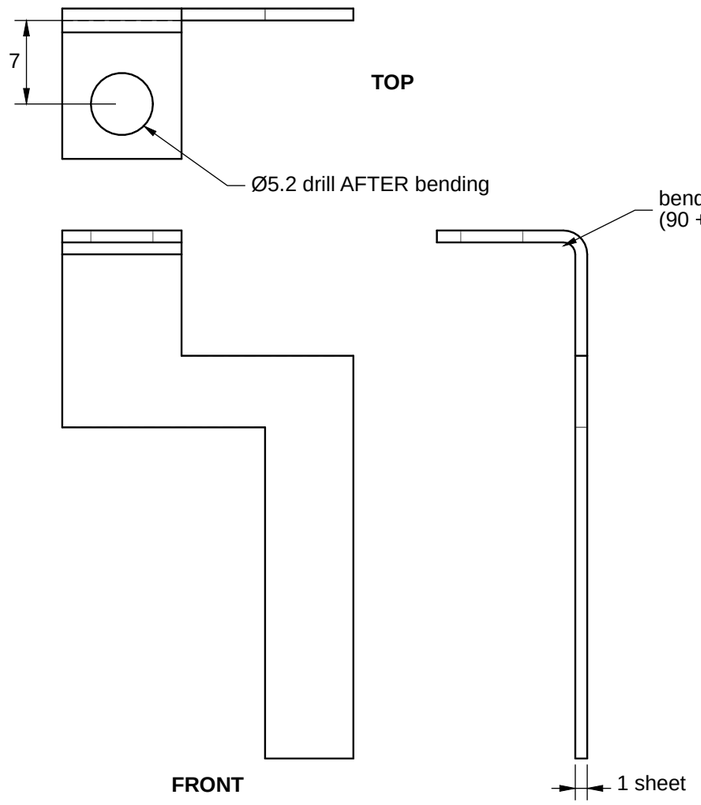
Constant	Value	Meaning
SLAB_T	6.6	c at pick contact = top of the stack: one slab (pose pick_bottom) = SLAB_T; n slabs = n x SLAB_T, nominal
CUP_HANG	2	the free cup hangs this far below c; the probe trips after this much buffer travel
PROBE_TRIP_TOL	0.5	trip tolerance on CUP_HANG (BUILD_GUIDE 7.6)
PROBE_WINDOW	12	max probing travel past the predicted contact
BUFFER_STROKE	20	spring buffer stroke (SMC ZPT25UUK20-B5-A10)
STATION_LENS_Z	23	station lens front above the pad top
STATION_READ_C	171.6	c at the station read = STATION_LENS_Z + CAM_TO_SLAB + SLAB_T + CUP_HANG
CAM_TO_SLAB	140	both cameras: lens front to the slab face
HEAD_CAM_DY	-5	head lens axis, Y from the cup axis (negative = toward the label end)
POST_H	218	guide-rod tops above the pad top
C_SAFE_LOADED	248	carry height, loaded and empty (C_SAFE_EMPTY is the same)
CARRY_SENSOR_C0	246.5	carry sensor: beam blocked from this c up to the top stop
CARRY_BAND	1	carry sensor band: within it either reading is allowed
C_MAX	250	top of the software's Z travel
Z_SWITCH_C	258	Z top (home) switch trip
C_HARD_TOP	262	top hard stop: z_plate meets z_motor_mount
C_MIN	-12	lowest c the design allows
C_HARD_BOTTOM	-14	bottom hard stop: Z carriage skirts on the z_end_cap lips
Z_CARRIAGE_ABOVE_CUP	295	Z carriage bottom edge above the cup face
Z_SPINE_BOTTOM	275	spine bottom end above the pad top (set +/-1, BUILD_GUIDE 7.1)
Z_SCREW_DX	24	lead screw axis, X from the cup axis
Z_SPINE_OFFSET_X	-10	Z spine centre, X from the cup axis

SlabRail - graded-slab batch scanner	**SHEET D-05 of D-17**
Z axis and head stack-up	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

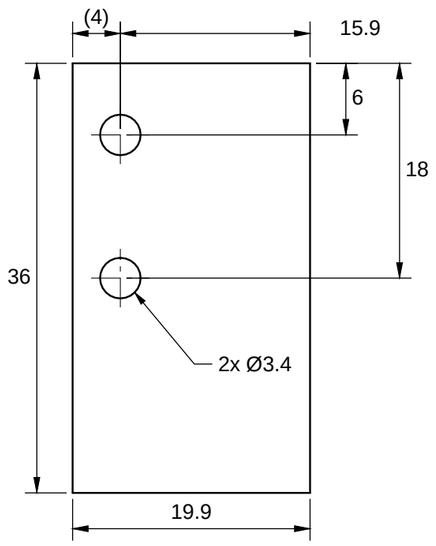
PROBE FLAG BLANK (2:1), flat, before bending
u from the neck centre line (= hole line), v from the bend line



PROBE FLAG AS BENT (2:1), for reference
(head frame: X along the row, Z above the cup face)



CARRY FLAG (2:1), flat
(seen from the front; top edge = trip edge)



Probe flag (BUILD_GUIDE 7.6; build.py probe_flag_blank.svg notes)

Aluminium sheet, 1 thick, a soft alloy: 1050, 3003 or 5052-H32, not 6061-T6 (a 1 mm inside bend radius can crack it [Likely], BOM.csv). Blank 24.4 x 55.25 (cut_list C-07 and BUILD_GUIDE 4.1 round the length to 55). 1. Cut and file to the outline; the blade's top edge is the trip edge: straight, square. 2. This view shows the face toward the buffer. Neck in the vise, jaws at the BEND line; bend the tab 90° TOWARD YOU. 3. THEN drill on the centre line, 7 from the plate's inner face. 4. Trim the tab to 2 past the hole. 5. Deburr: it rides in head_foot's guide slot and the sensor slot. Outline: cad/out/probe_flag_blank.dxf (1:1 print: .svg).

Carry flag (BUILD_GUIDE 7.4a)

Aluminium sheet, 1 thick, flat: 1050, 3003 or 5052-H32 as the probe flag (BOM.csv) (cut_list C-10). File the top edge straight. Fix to carry_flag_arm's leg end with 2 x M3x8, inner edge toward the spine, top edge flush with the arm's top.

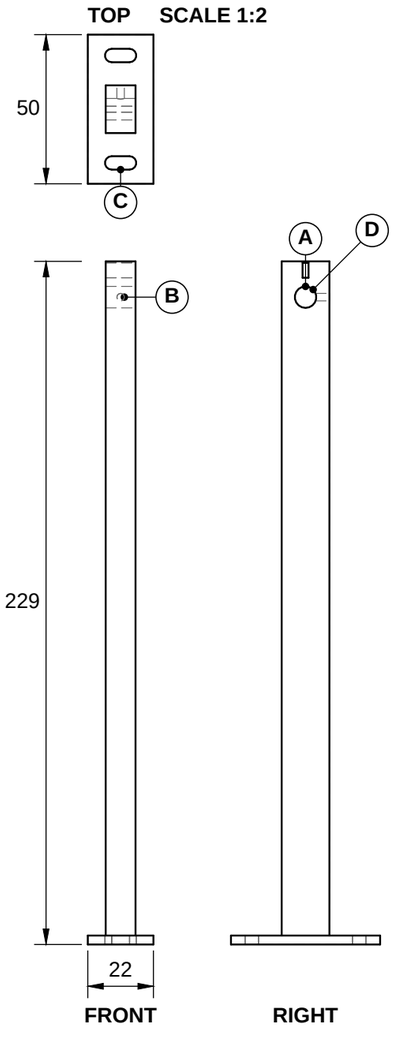
Tolerance: outlines PROPOSED +/-0.2 (no doc states one); bend 90 +/-0.5° (BUILD_GUIDE 7.6). The trip points are set by adjustment on assembly (probe 7.6, carry 7.4a), not by these outlines.

SlabRail - graded-slab batch scanner	**SHEET D-06 of D-17**
Flat parts (probe flag, carry flag)	SCALE 2:1
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

gate_post_L

Printed part (P), qty 1, cad/out/printed/gate_post_L.step. Print: PETG, upright on its foot (as exported), 10 mm brim, 4 walls, 30% infill; needs 230 mm Z height, else print lying on its side with supports

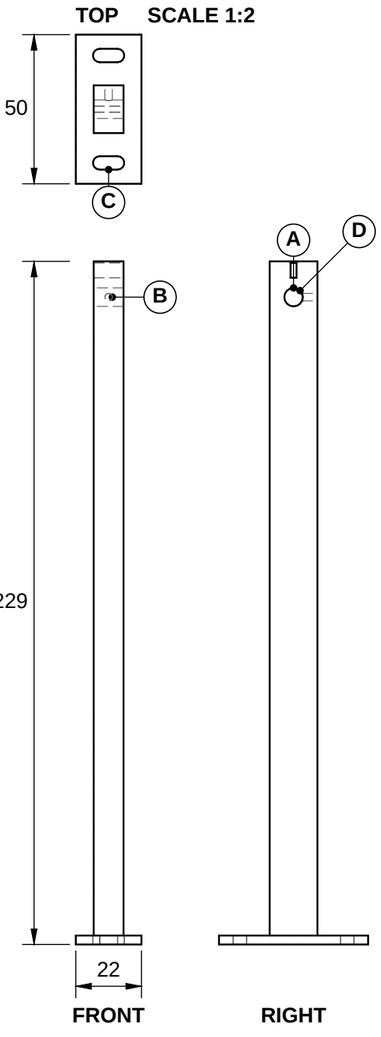
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø7.2 THRU, axis X: laser barrel bore (7 mm Class 2 laser)
B	Ø2.5 from the +Y face into the barrel bore, axis Y: M3 set screw onto the barrel (tap or self-form)
C	2x slot 4.5 wide x 10.5 long, axis Z, centres 36 apart in Y: yaw slot, wood screws
D	Barrel bore axis 217 above the foot underside: on bare plywood the beam is GATE_Z = 214 above the pad top (BUILD_GUIDE 5.9: no shim)



gate_post_R

Printed part (P), qty 1, cad/out/printed/gate_post_R.step. Print: PETG, upright on its foot (as exported), 10 mm brim, 4 walls, 30% infill; needs 230 mm Z height, else print lying on its side with supports

	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø6.2 THRU, axis X: receiver barrel bore (6 mm module)
B	Ø2.5 from the +Y face into the barrel bore, axis Y: M3 set screw onto the barrel (tap or self-form)
C	2x slot 4.5 wide x 10.5 long, axis Z, centres 36 apart in Y: yaw slot, wood screws
D	Barrel bore axis 217 above the foot underside: on bare plywood the beam is GATE_Z = 214 above the pad top (BUILD_GUIDE 5.9: no shim)

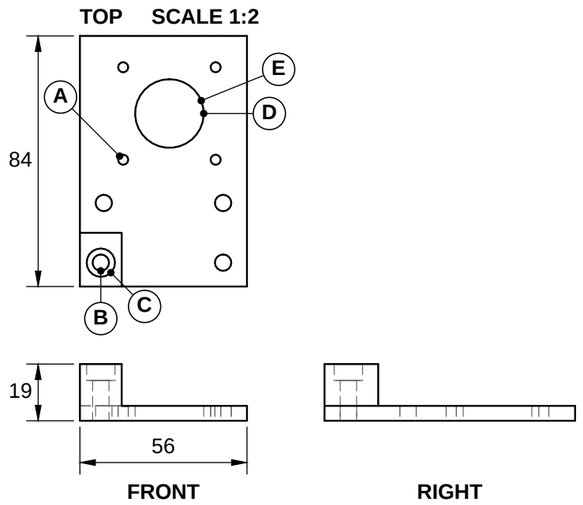


Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-07 of D-17**
Printed parts: gate posts	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

x_motor_mount

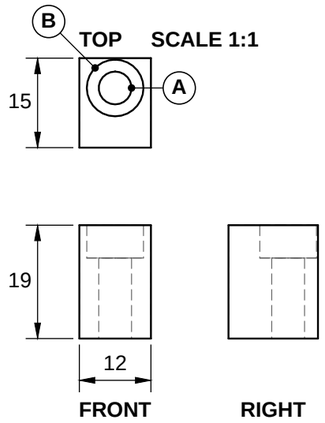
Printed part (P), qty 1, cad/out/printed/x_motor_mount.step. Print: PETG, beam face down (as exported), 5 walls, 40% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	4x Ø3.4 THRU, axis Z, on 31 x 31 (X x Y): M3 clearance
B	4x Ø5.5 THRU, axis Z, centres spread over 41 x 20 (X x Y), not a grid: M5 clearance
C	Ø9.5 x 5.5 deep, axis Z: M5 head, sunk: 1x, at the stop-end M5 hole only
D	Ø23 THRU, axis Z: NEMA 17 pilot clearance
E	Left end = right X hard stop face, 30 from the motor axis; motor axis 20 above the beam top

x_bumper_L

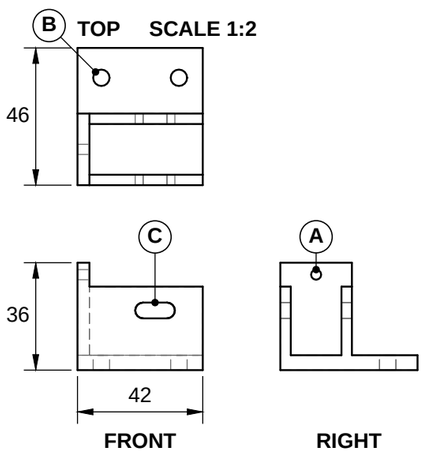
Printed part (P), qty 1, cad/out/printed/x_bumper_L.step. Print: PETG, beam face down (as exported), 4 walls, 40% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU, axis Z: M5 clearance
B	Ø9.5 x 5.5 deep, axis Z: M5 head, sunk

x_idler_mount

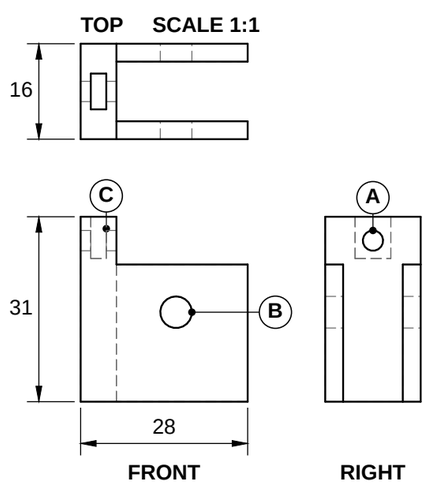
Printed part (P), qty 1, cad/out/printed/x_idler_mount.step. Print: PETG, foot down, 5 walls, 40% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø3.4 THRU, axis X: M3 clearance
B	2x Ø5.5 THRU, axis Z, 26 apart in X: M5 clearance
C	2x slot 5.3 wide x 13.3 long, axis Y: M5 idler axle (tension travel)

x_idler_slider

Printed part (P), qty 1, cad/out/printed/x_idler_slider.step. Print: PETG, as exported, 5 walls, 50% infill



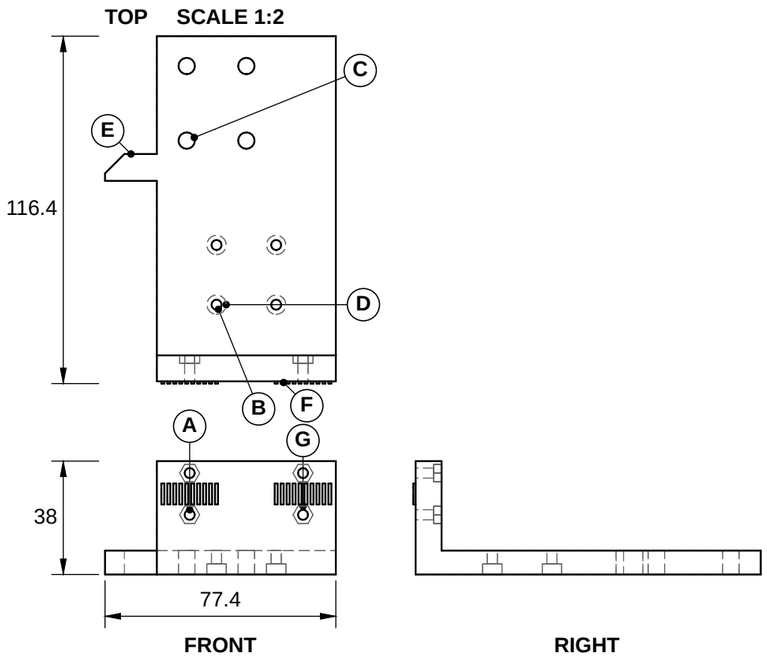
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø3.4 THRU (2 walls on one axis), axis X: M3 clearance
B	Ø5.3 THRU (2 walls on one axis), axis Y: M5 axle
C	M3 nut slot 2.6 wide, from the top, in the bridge

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-08 of D-17**
Printed parts: X drive	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

x_bracket

Printed part (P), qty 1, cad/out/printed/x_bracket.step. Print: PETG, print flat on its spine face (as exported), 6 walls, 50% infill; the finger off its left edge works the X home switch



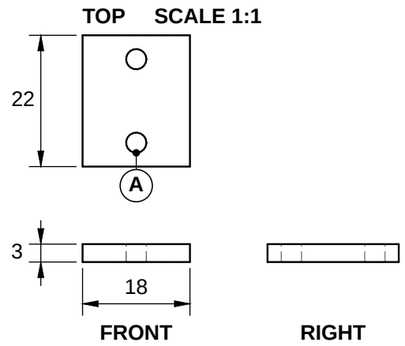
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	4x Ø3.4 THRU, axis Y, on 38 x 14 (X x Z): belt_clamp caps, 2 x M3x12 + nut per cap (BUILD_GUIDE 6.5)
B	4x Ø3.4 THRU, axis Z, on 20 x 20 (X x Y): M3 clearance
C	4x Ø5.5 THRU, axis Z, on 20 x 25 (X x Y): M5 clearance
D	4x Ø6.5 x 3.5 deep, axis Z, on 20 x 20 (X x Y): M3 head counterbore
E	X-switch finger 17.4 past the left edge; flat underside at machine Z 369.5; 45° lead-in 6.5 deep
F	GT2 rack on the tab floor, both sides: teeth at 2 pitch (belt teeth down, belt_clamp cap on top)
G	4x M3 nut trap, hex 5.8 across flats x 2.6 deep, in the tab underside under the belt_clamp holes

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

belt_clamp

Printed part (P), qty 2, cad/out/printed/belt_clamp.step. Print: PETG, flat, 100% infill

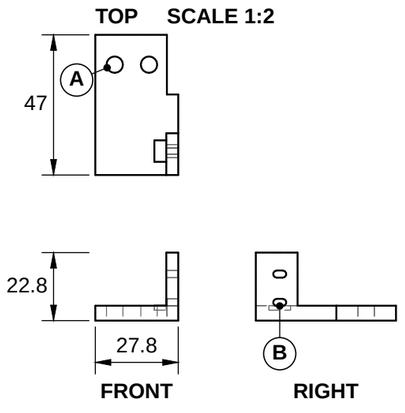
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø3.4 THRU, axis Z, 14 apart in Y: M3 clearance



x_switch_mount

Printed part (P), qty 1, cad/out/printed/x_switch_mount.step. Print: PETG, front plate down (as exported), 4 walls, 40% infill; X home switch on 2 x M2x12, 2 x M5x10 button heads

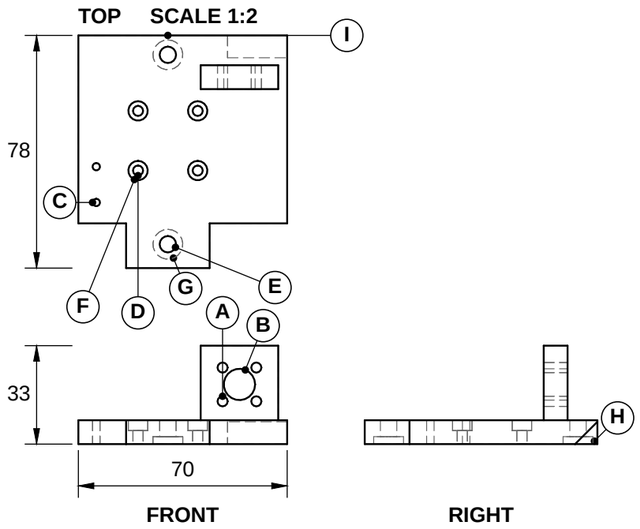
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø5.5 THRU, axis Z, 11.5 apart in X: M5 clearance
B	2x slot 2.4 wide x 4.4 long, axis X, centres 9.5 apart in Z: M2



SlabRail - graded-slab batch scanner	**SHEET D-09 of D-17**
Printed parts: X carriage	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

z_plate

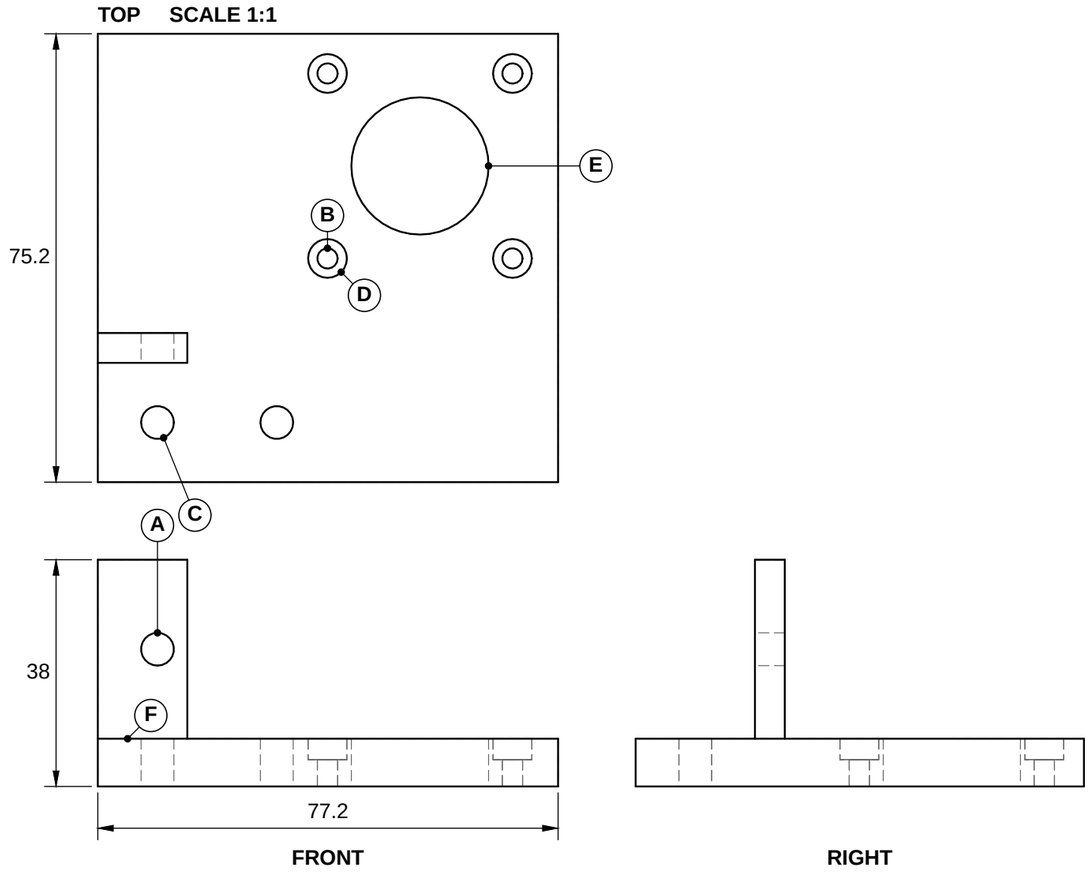
Printed part (P), qty 1, cad/out/printed/z_plate.step. Print: PETG, carriage face down, nut shelf up, 6 walls, 50% infill; the switch chamfer at the top right starts at the bed (45 deg)



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	4x Ø3.4 THRU, axis Y, on 11.3 x 11.3 (X x Z): M3 clearance
B	Ø10.5 THRU, axis Y: T8 nut body clearance
C	2x Ø2.5 THRU, axis Z, 12 apart in Y: carry_flag_arm, M3x10 self-tapping
D	4x Ø3.4 THRU, axis Z, on 20 x 20 (X x Y): M3 clearance
E	2x Ø5.5 THRU, axis Z, 63.5 apart in Y: M5 clearance
F	4x Ø6.5 x 3.5 deep, axis Z, on 20 x 20 (X x Y): M3 head counterbore
G	2x Ø10 x 2.5 deep, axis Z, 63.5 apart in Y: M5 button-head recess
H	Z-switch chamfer 45°, 7.5 x 7.5, 20 long at the right end of the top back edge
I	Top edge (Z top hard stop) 58 and T8 flange shelf top 48 above the Z carriage's bottom edge

z_motor_mount

Printed part (P), qty 1, cad/out/printed/z_motor_mount.step. Print: PETG, motor face down, tab up, 5 walls, 40% infill



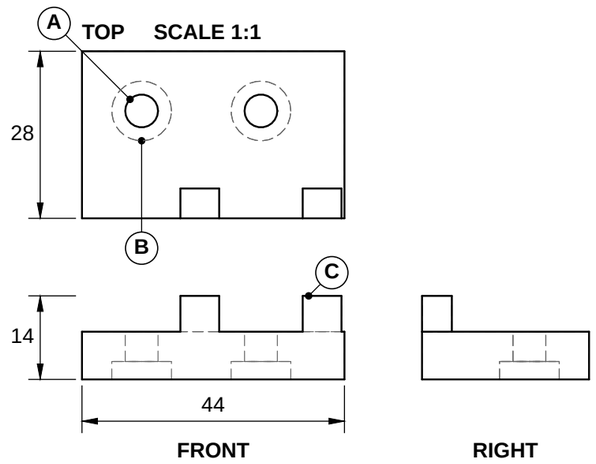
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU, axis Y: M5 clearance
B	4x Ø3.4 THRU, axis Z, on 31 x 31 (X x Y): M3 clearance
C	2x Ø5.5 THRU, axis Z, 20 apart in X: M5 clearance
D	4x Ø6.5 x 3.5 deep, axis Z, on 31 x 31 (X x Y): M3 head counterbore
E	Ø23 THRU, axis Z: NEMA 17 pilot clearance
F	Plate underside sits on the spine top and is the Z top hard stop; tab hole 15 below it (left slot)

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-10 of D-17**
Printed parts: Z carriage plate, Z motor mount	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

z_end_cap

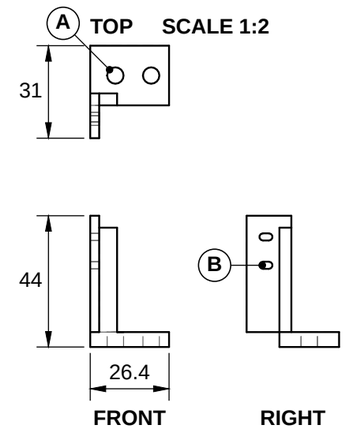
Printed part (P), qty 1, cad/out/printed/z_end_cap.step. Print: PETG, as exported (lips up), 5 walls, 50% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø5.5 THRU, axis Z, 20 apart in X: M5 clearance
B	2x Ø10 x 3 deep, axis Z, 20 apart in X: M5 button-head recess
C	Two lips 6 above the plate top, under the Z carriage skirts: the Z bottom hard stop (c = C_HARD_BOTTOM)

z_switch_mount

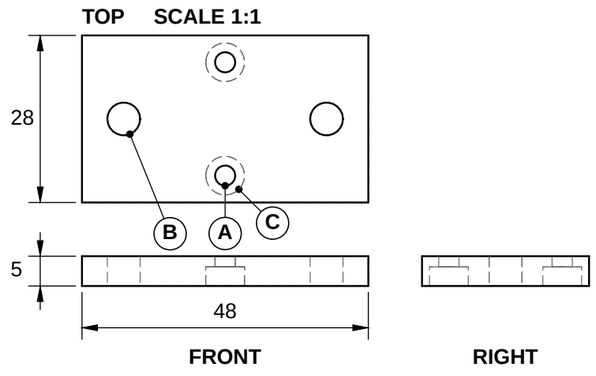
Printed part (P), qty 1, cad/out/printed/z_switch_mount.step. Print: PETG, spine plate down (as exported), 4 walls, 40% infill; Z top switch on 2 x M2x12, 2 x M5x10 button heads



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø5.5 THRU, axis Z, 12 apart in X: M5 clearance
B	2x slot 2.4 wide x 4.4 long, axis X, centres 9.5 apart in Z: M2

carry_sensor_mount

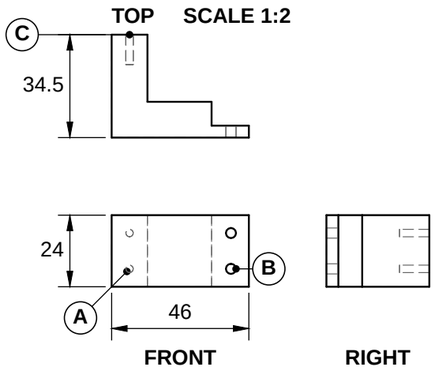
Printed part (P), qty 1, cad/out/printed/carry_sensor_mount.step. Print: PETG, flat on its spine face (as exported), 4 walls, 40% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø3.4 THRU, axis Z, 19 apart in Y: M3 clearance
B	2x Ø5.5 THRU, axis Z, 34 apart in X: M5 clearance
C	2x Ø6.5 x 3.2 deep, axis Z, 19 apart in Y: M3 head counterbore

carry_flag_arm

Printed part (P), qty 1, cad/out/printed/carry_flag_arm.step. Print: PETG, standing on its bottom face (as exported), 4 walls, 40% infill



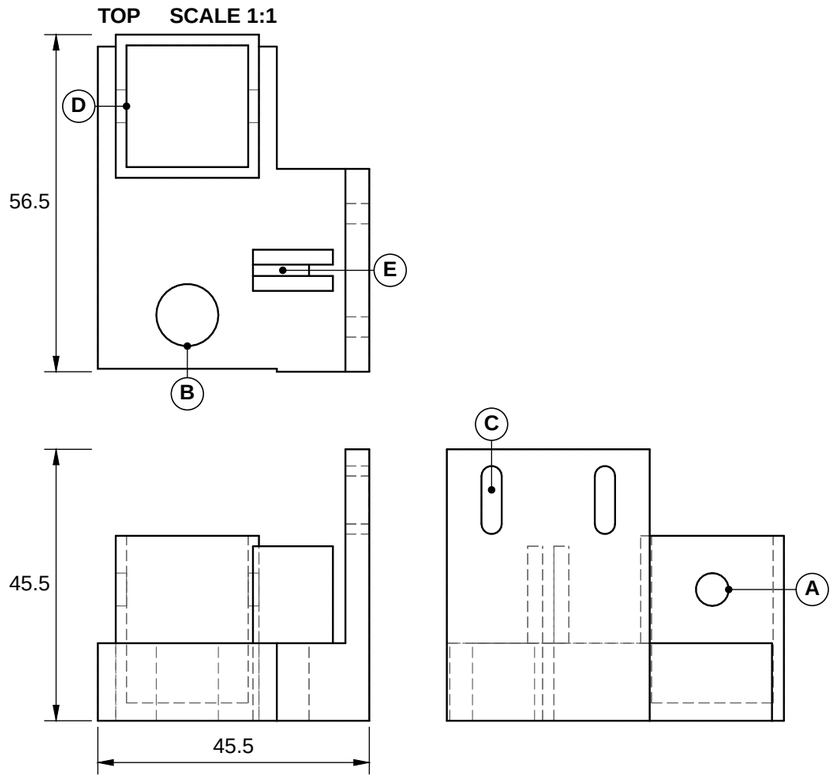
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø2.5 x 10 deep, axis Y, 12 apart in Z: carry flag, M3x8 self-tapping
B	2x Ø3.4 THRU, axis Y, 12 apart in Z: M3 clearance
C	Leg end face (carry flag seat) 30.5 behind the ear face that sits on the z_plate

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-11 of D-17**
Printed parts: Z end cap, Z switch mount, carry-height sensor	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

head_foot

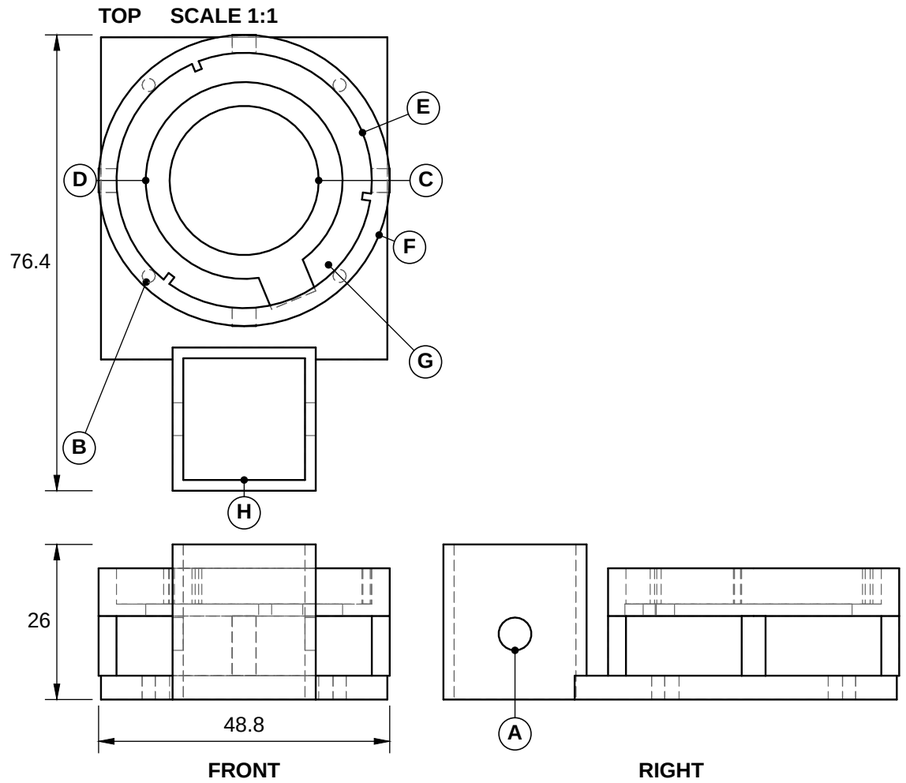
Printed part (P), qty 1, cad/out/printed/head_foot.step. Print: PETG, foot down (as exported), 5 walls, 40% infill; clear the 1.9 mm flag slot and the board slots with a hacksaw blade



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU (2 walls on one axis), axis X: M5 clearance
B	Ø10.4 THRU, axis Z: M10x1 buffer body
C	2x slot 3.4 wide x 11.4 long, axis X, centres 19 apart in Y: endstop board M3 slots (trip adjustment)
D	Drop-arm socket 20.4 square, floor 3 above the underside; cross hole 19 above the floor (match-drill the arm)
E	Flag guide slot 1.9 wide through the fins and the foot (clear it after printing); underside at c + 62

camera_bracket

Printed part (P), qty 1, cad/out/printed/camera_bracket.step. Print: PETG matte black, shelf face down (as exported), 4 walls, 40% infill, brim; the ring seat bridges ~30 mm between its 4 legs (tree supports there if your printer sags)



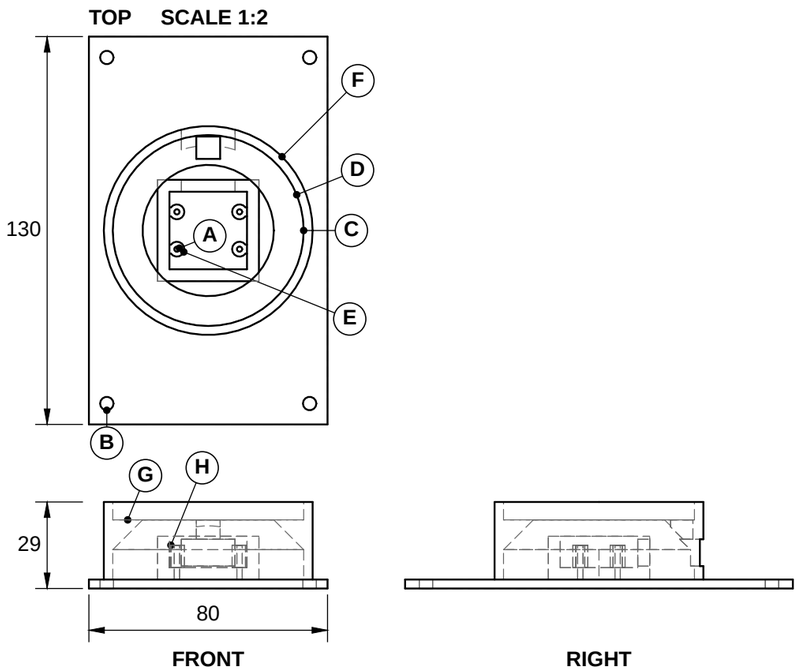
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU (2 walls on one axis), axis X: M5 clearance
B	4x Ø2.2 THRU, axis Z, on 32 x 32 (X x Y): ELP board, M2 + nuts, about the lens axis ([Guessing], 3.3)
C	Ø25 THRU, axis Z: lens window
D	Ø33 THRU, interrupted, axis Z: ring lip (ring back face rests on it)
E	Ø42.8 x 6 deep, interrupted, axis Z: head LED ring seat bore (ribbed)
F	Ø48.8 x 8 long (outside), axis Z: ring seat tube
G	Ring seat: 3 ribs leave Ø40 (shave for a larger ring, BUILD_GUIDE 3.3); lip 16 below the board shelf
H	Lens axis 40 from the drop-arm centre (ARM_TO_CUP + lens offset HEAD_CAM_DY = -5)

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-12 of D-17**
Printed parts: head	SCALE 1:1
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

station_mount

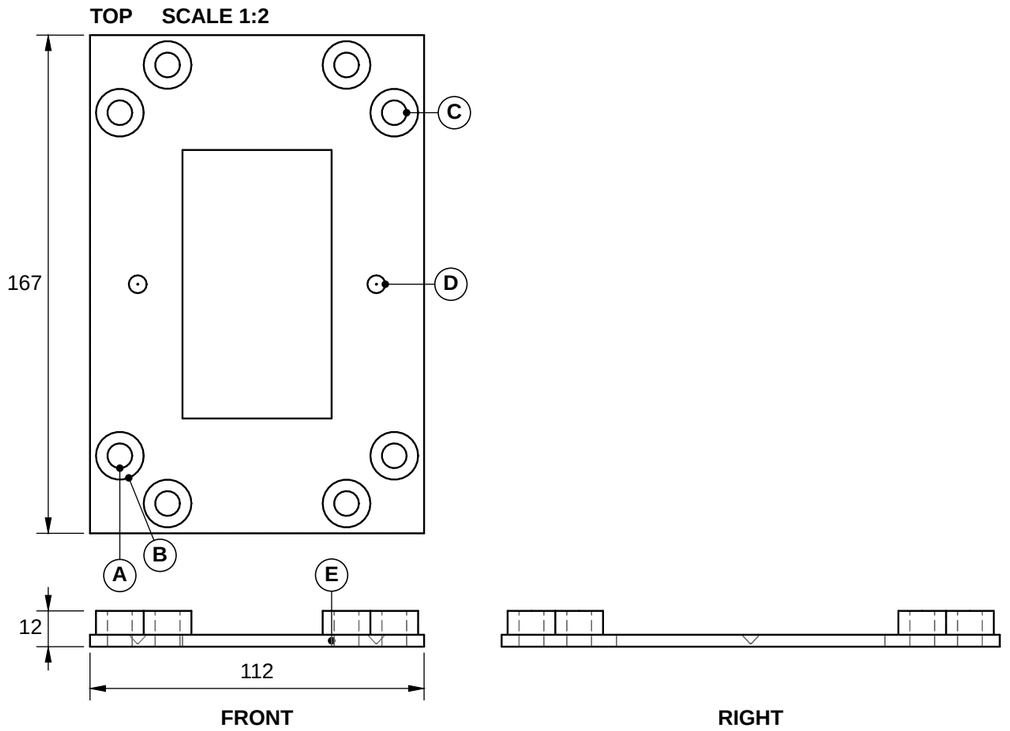
Printed part (P), qty 1, cad/out/printed/station_mount.step. Print: PETG black, 3 walls, 25% infill; ribbon slot bridges



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	4x Ø1.8 x 11.5 deep, axis Z, on 21 x 12.5 (X x Y): camera board, M2 self-tapping
B	4x Ø4.5 THRU, axis Z, on 68 x 116 (X x Y): #8 wood screws into the bed
C	Ø64 x 10 deep, axis Z: hollow under the ring shelf (45° cone roof)
D	Ø64 x 6 deep, axis Z: station LED ring seat bore
E	4x Ø5 x 7.5 long (outside), part-round (joins the wall), axis Z: M2 boss
F	Ø70 x 26 long (outside), axis Z: ring seat tube
G	Ring shelf (ring back face, LEDs up) 20 above the base-plate top, which is level with the pad tops
H	Camera bosses: tops 11.5 above the base-plate top (board back); lens front STATION_LENS_Z = 23

drill_template

Printed part (P), qty 1, cad/out/printed/drill_template.step. Print: PETG (or PLA), plate down (as exported), 0.2 mm, 4 walls, 50% infill, bosses solid (build tool)



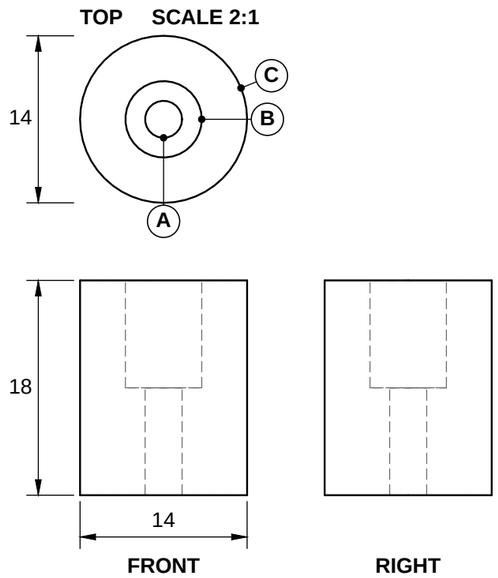
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	8x Ø8.3 THRU, axis Z, centres spread over 92 x 147 (X x Y), not a grid: dowel drill guide
B	8x Ø16 x 8 long (outside), axis Z: drill-guide boss
C	Side-rod holes 92 x 115; front/back-rod holes 60 x 147 (centres)
D	Two cone dimples Ø6, 80 apart on the row centre line (the awl finds them)
E	Window 50 x 90 to see the pencil centre line

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-13 of D-17**
Printed parts: station mount, drill template	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

awl_adapter

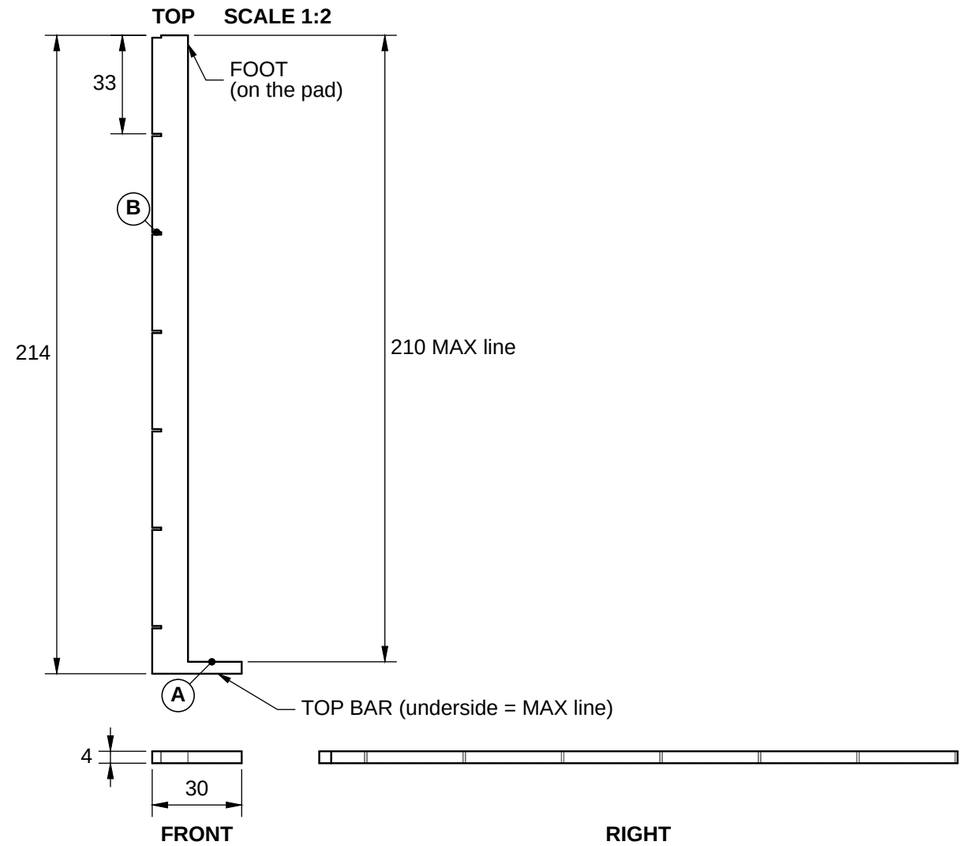
Printed part (P), qty 1, cad/out/printed/awl_adapter.step. Print: PETG, as exported, 100% infill (build tool) + M5 heat-set insert + 3 mm steel pin



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø3.1 THRU, axis Z: 3 mm steel pin, pressed in + CA
B	Ø6.4 x 9 deep, axis Z: M5 heat-set insert pocket
C	Ø14 x 18 long (outside), axis Z: body

fill_gauge

Printed part (P), qty 2, cad/out/printed/fill_gauge.step. Print: PLA, flat (as exported), 3 walls, 20% infill (loading tool)



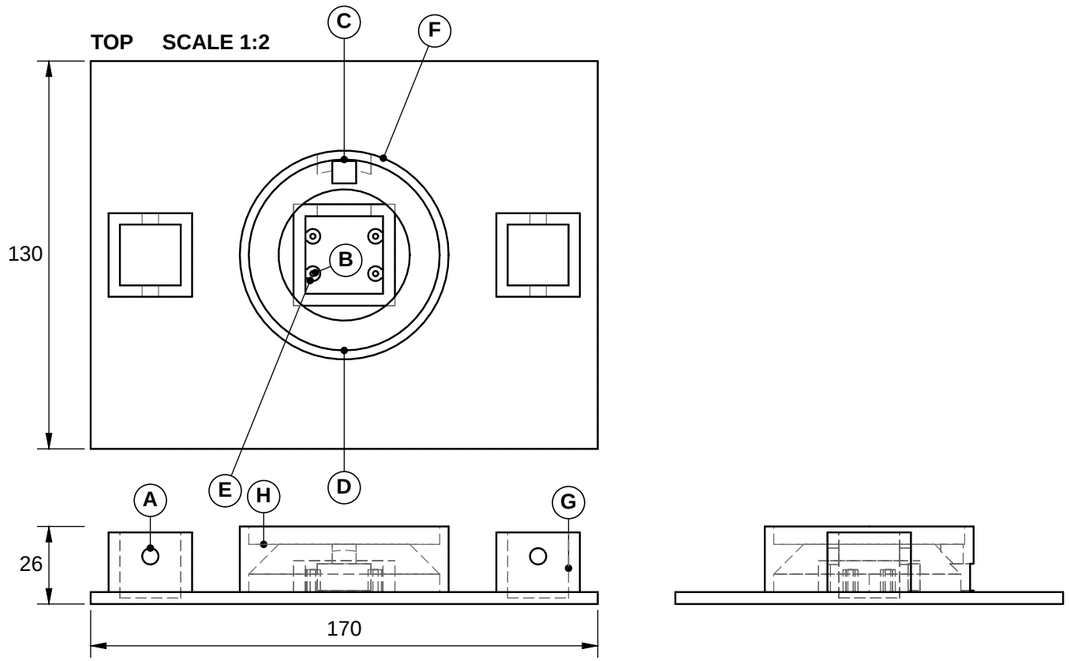
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	MAX line = underside of the top bar, 210 above the foot (MAX_LINE = 210)
B	Count notches every 33 (5 slabs), 0.8 wide, 3 deep

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-14 of D-17**
Printed parts: awl adapter, fill gauge	SCALE 2:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

phase0_base

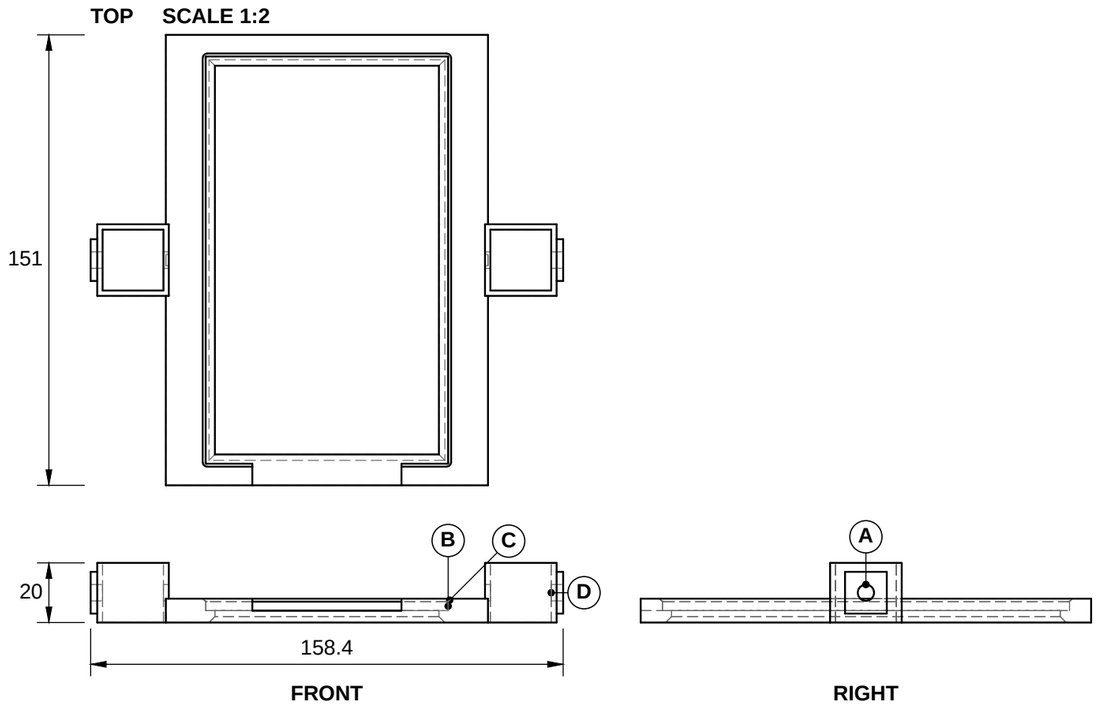
Printed part (P), qty 1, cad/out/printed/phase0_base.step. Print: PLA/PETG black (Phase 0 test jig), plate down (as exported), 3 walls, 25% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø5.5 THRU (2 walls on one axis), axis Y, 130 apart in X: M5 clearance
B	4x Ø1.8 x 7.5 deep, axis Z, on 21 x 12.5 (X x Y): camera board, M2 self-tapping
C	Ø64 x 6 deep, axis Z: hollow under the ring shelf (45° cone roof)
D	Ø64 x 6 deep, axis Z: station LED ring seat bore
E	4x Ø5 x 7.5 long (outside), part-round (joins the wall), axis Z: M2 boss
F	Ø70 x 22 long (outside), axis Z: ring seat tube
G	Post sockets 20.4 square, floor 2 above the underside, centres 130 apart
H	Ring shelf 20 and socket cross holes 16 above the underside

phase0_tray

Printed part (P), qty 1, cad/out/printed/phase0_tray.step. Print: PLA/PETG (Phase 0 test jig), ledges down (as exported), 3 walls, 25% infill



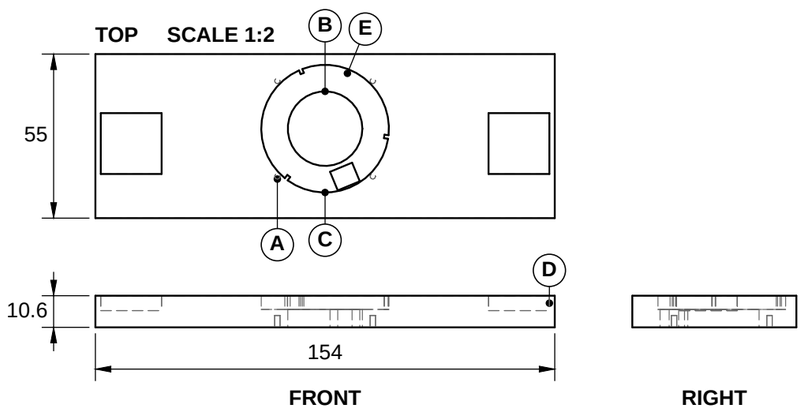
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	Ø5.5 THRU (4 walls on one axis), axis X: M5 clearance
B	Slab recess 81.3 x 136.5, 4 deep; opening 75.1 x 130.3; the ledges carry the slab's front face
C	45° lead-in round the top edge of the recess, 1 wide x 1 deep
D	Post clamps 20.4 square, centres 130 apart

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

SlabRail - graded-slab batch scanner	**SHEET D-15 of D-17**
Printed parts: Phase 0 jig (base, tray)	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

phase0_top

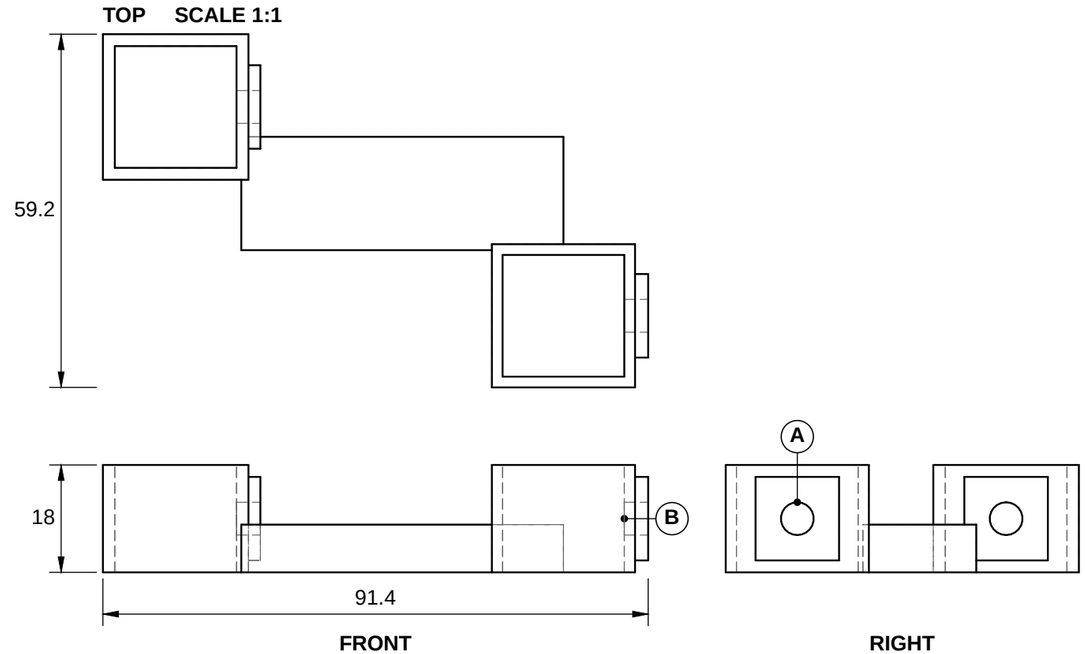
Printed part (P), qty 1, cad/out/printed/phase0_top.step. Print: PLA/PETG (Phase 0 test jig), ring pocket up (as exported), 3 walls, 25% infill



	FEATURES (axes = the STEP file's; positions: see the STEP)
A	4x Ø1.8 x 4 deep, axis Z, on 32 x 32 (X x Y): M2 standoffs, blind
B	Ø25 THRU, axis Z: lens window
C	Ø42.8 x 4.6 deep, interrupted, axis Z: head LED ring pocket (ribbed)
D	Blind post sockets 20.4 square, 5 deep; bar runs 30 toward the label end and 25 toward the back of the socket centre line
E	Head-ring pocket ribs leave Ø40 (as camera_bracket's seat)

phase0_occluder_mount

Printed part (P), qty 1, cad/out/printed/phase0_occluder_mount.step. Print: PLA/PETG (Phase 0 test jig), flat on its beam (as exported), 4 walls, 30% infill



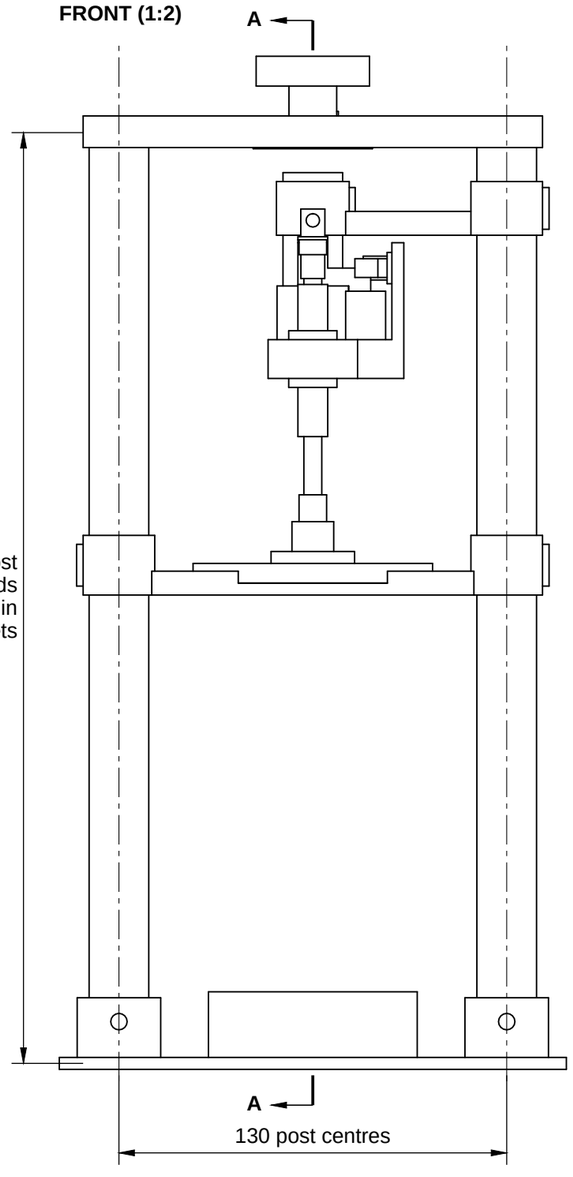
	FEATURES (axes = the STEP file's; positions: see the STEP)
A	2x Ø5.5 THRU, axis X, 35 apart in Y: M5 clearance
B	Post bore 20.4 square and stub bore 20.4 square, centres 65 apart in X and 35 in Y

Printed parts are drawn as exported (cad/out/printed/<part>.step): print orientation, bounding-box minimum at the origin; FRONT looks along +Y, TOP down -Z, RIGHT along -X. Dimensions measured on that shape. Drill M3 holes 3.4 and M5 holes 5.5 after printing (BUILD_GUIDE 3.4). Tolerance: see D-00 (PROPOSED).

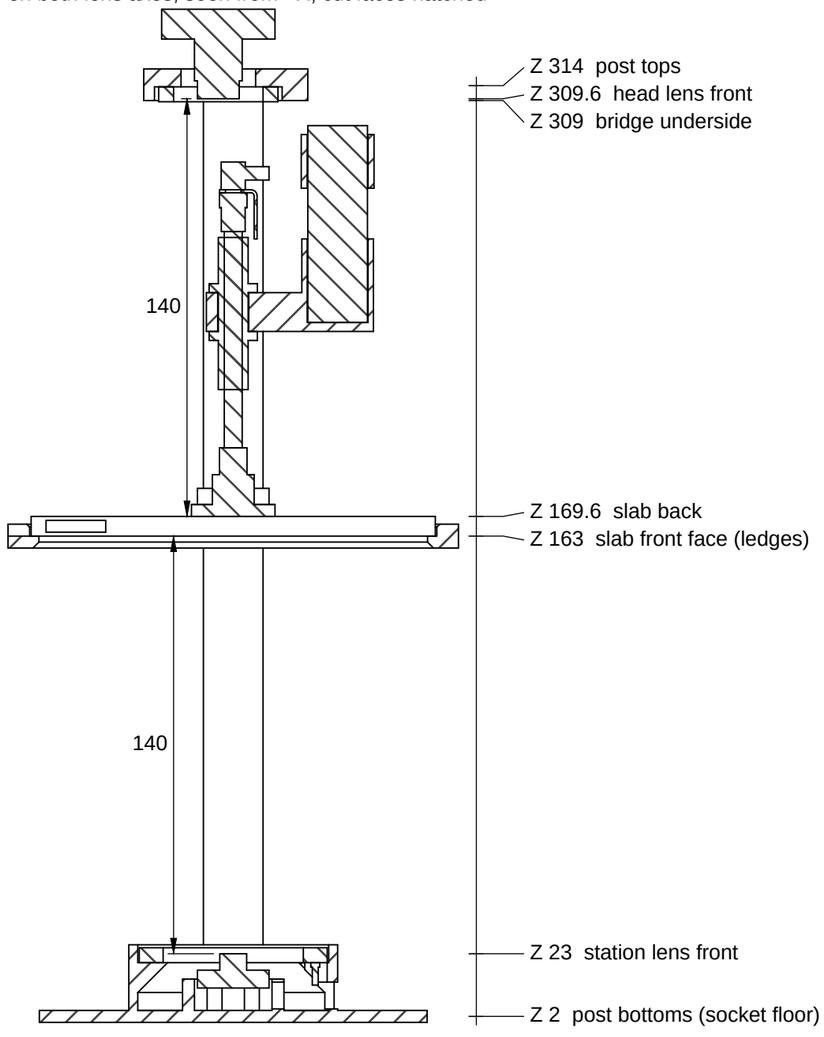
SlabRail - graded-slab batch scanner	**SHEET D-16 of D-17**
Printed parts: Phase 0 jig (top, occluder mount)	SCALE 1:1, 1:2 (as noted)
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**

PHASE 0A TEST JIG: one slab between the two cameras, as on the machine (BUILD_GUIDE 1.5); fasteners not shown

FRONT (1:2)

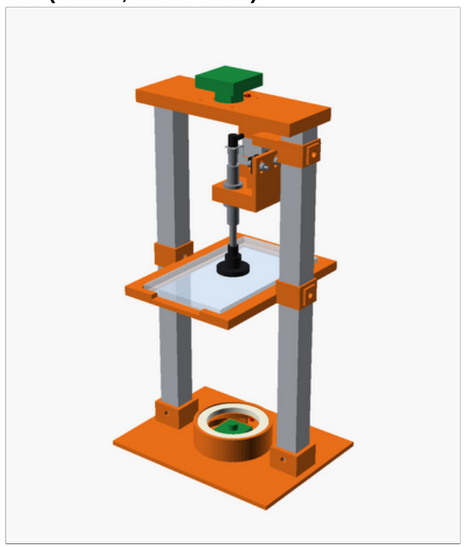


SECTION A-A (1:2)
on both lens axes, seen from +X; cut faces hatched



Heights above the base underside

ISO (render; not to scale)



Part	Type	Qty	Sheet
phase0_base (camera mast, ring seat, post sockets)	P	1	D-15
phase0_tray (the slab on edge ledges)	P	1	D-15
phase0_top (bridge: head camera, head ring)	P	1	D-16
phase0_occluder_mount	P	1	D-16
head_foot, the machine's own (occluder)	P	1	D-12
2020 post, 312, cut square +/-0.5 (C-08)	C	2	D-02
2020 arm stub, 66 (C-09)	C	1	D-02
probe flag (C-07)	S	1	D-06
cameras, LED rings, SMC buffer + cup + elbow, endstop board	B	-	BOM.csv

The posts set the bridge height: cut them square. Both lens-to-slab distances (SECTION A-A) equal CAM_TO_SLAB = 140 on the machine (set +/-2, BUILD_GUIDE 1.5). Phase 0 is the go / no-go test before any motion parts are bought.

SlabRail - graded-slab batch scanner	**SHEET D-17 of D-17**
Phase 0 test jig	SCALE 1:2
Rev A - DRAFT FOR DESIGN REVIEW, not released for fabrication	UNITS: mm 3RD ANGLE
SOURCE: cad/model.py @ last CAD change 3f36fc8	DATE 2026-10-27 **DO NOT SCALE**